TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

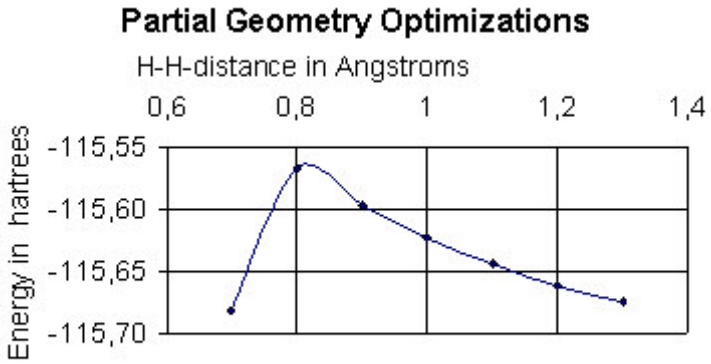
Prof. N. Rösch

Full Solution of Exercise 19

a. Eclipsed conformation is a saddle point.

b. A pre-filled form for the first point of the approximated reaction path has been already given at the first level of this solution.

c.



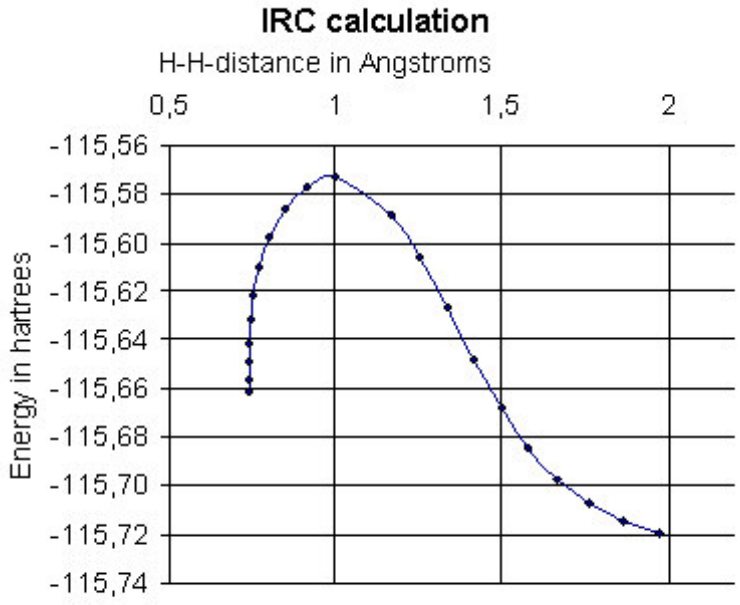
d. Reaction barrier

with consideration of the zero point energies of both the reactant and the transition state: 372 kJ/mol

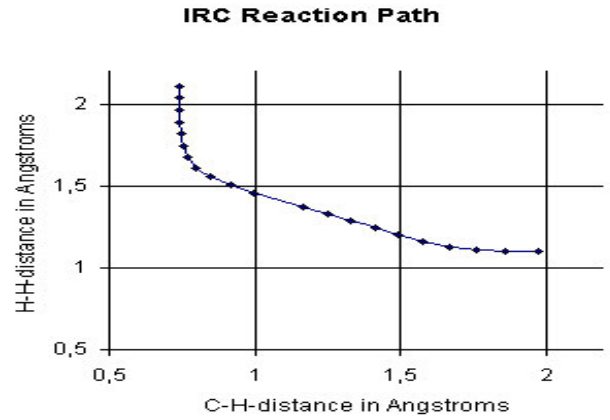
without consideration of the zero point energies: 396 kJ/mol

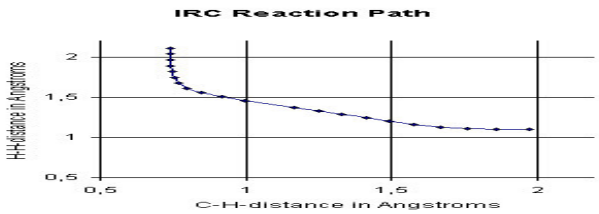
The consideration of the zero point energies lowers the reaction barrier because the imaginary frequency gives no contribution to the zero point energy of the transition state.

e. The following figure shows a plot of the potential energy vs. the distance between the two hydrogen atoms that form molecular hydrogen in the reaction.



In the next figure two parameters of the intrinsic reaction path are plotted (H-H-distance vs. C-H-distance of the separating methyl H atom).





[General help] [Overview] [Compute] [Contact the course supervisor]

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